

Electric Field

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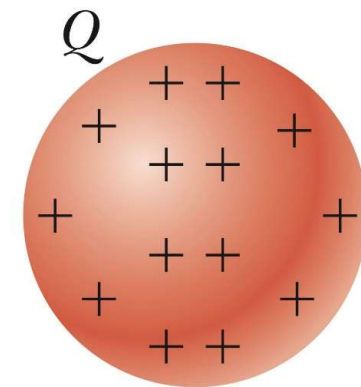
Electric Field – Definition

- ▶ When a point charge is placed at a certain point in space, it produces a **physical effect** at every point in the region of space around it, called the **electric field**.
- ▶ If a test charge is placed at a point in that space the electric field at that point acts with a force on the test charge.
- ▶ This field depends on the **source charge** and the **location** at which the **test** charge is placed and **does not depend on the test charge**

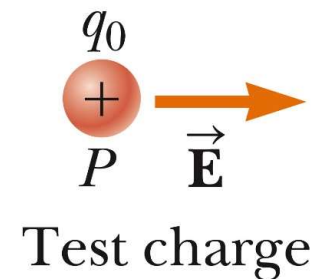
Electric Field – Definition, cont

- ▶ The electric field is defined as the electric force on the test charge per unit charge.
- ▶ The electric field vector, $\vec{E}$, at a point in space is defined as the electric force acting on a positive test charge, q_0 , placed at that point divided by the test charge:

$$\vec{E} = \frac{\vec{F}_e}{q_0}$$



Source charge



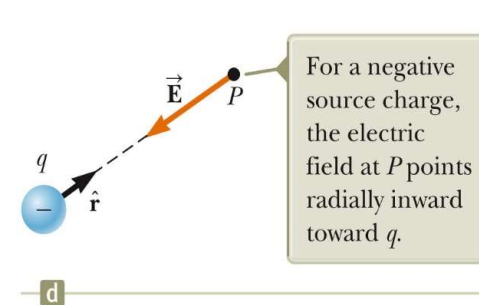
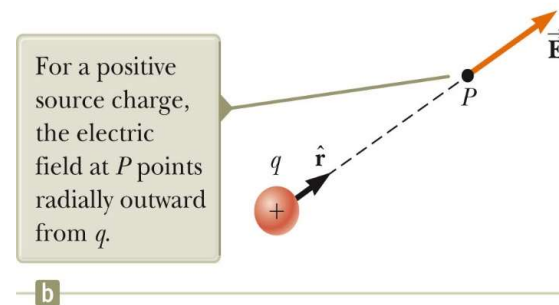
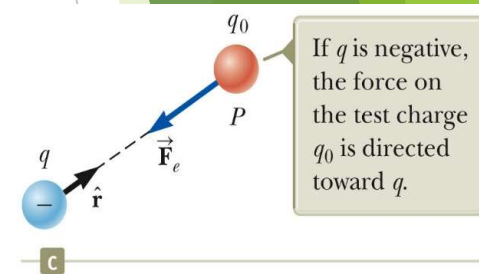
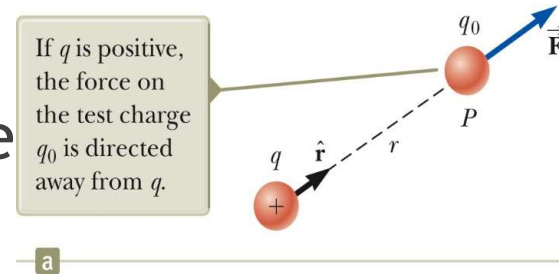
Test charge

Electric Field

- ▶ $\underline{E}$ is the field produced by some charge or charge distribution, separate from the test charge.
- ▶ The existence of an electric field is a property of the source charge.
 - ▶ The presence of the test charge is not necessary for the field to exist.
- ▶ The test charge serves as a detector of the field.
- ▶ The direction of $\underline{E}$ is that of the force on a positive test charge.
- ▶ The SI units of $\underline{E}$ are N/C.
- ▶ We can also say that an electric field exists at a point if a test charge at that point experiences an electric force.

Relationship Between F and E

- If q is positive, the force and the field are in the same direction.
- If q is negative, the force and the field are in opposite directions.
- b) The direction of the field is also away from the positive source charge.
- d) The field is also toward the negative source charge.



Electric Field, Point Charge

$$E = \frac{F}{q} = \frac{kqQ/r^2}{q}$$

$$E = k \frac{Q}{r^2};$$

[single point charge] **(21-4a)**

or, in terms of ϵ_0 as in Eq. 21-2 ($k = 1/4\pi\epsilon_0$):

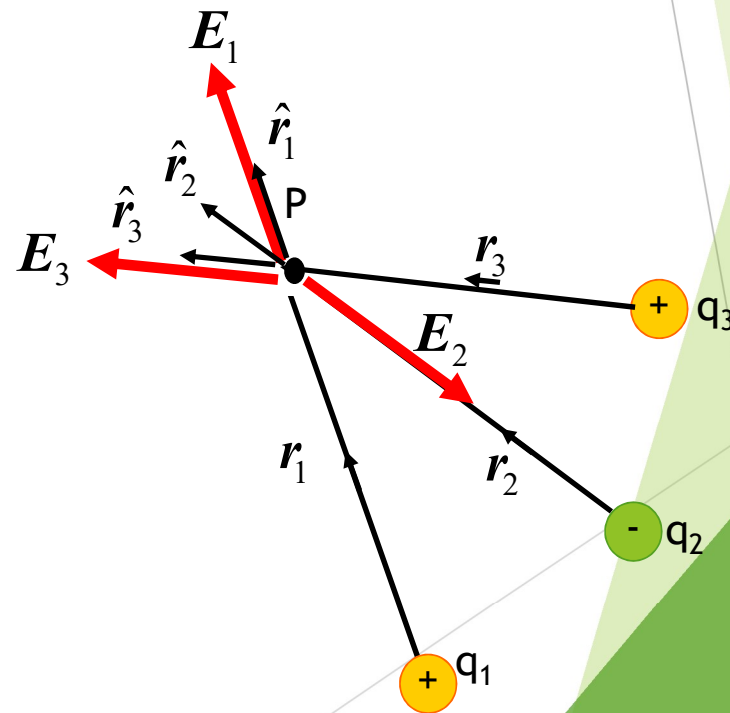
$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}.$$

[single point charge] **(21-4b)**

Superposition of Electric Fields

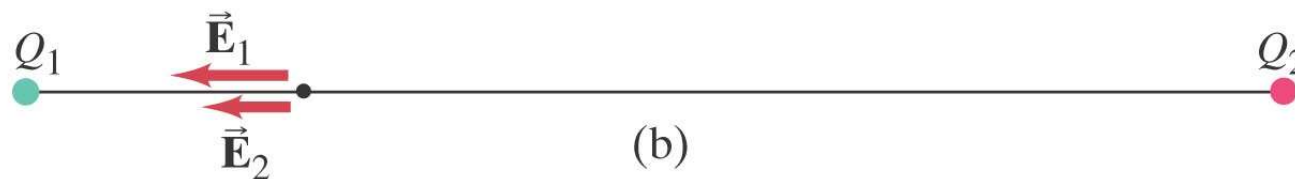
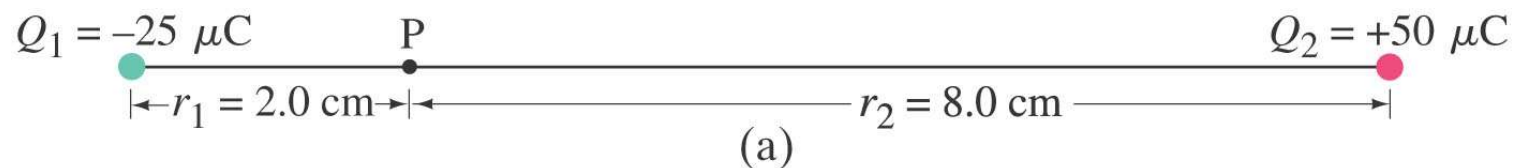
- At any point P , the total electric field due to a group of source charges equals the vector sum of the electric fields of all the charges.

$$\therefore \mathbf{E} = \sum_i \mathbf{E}_i = k_e \sum_i \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i$$



Example 1:

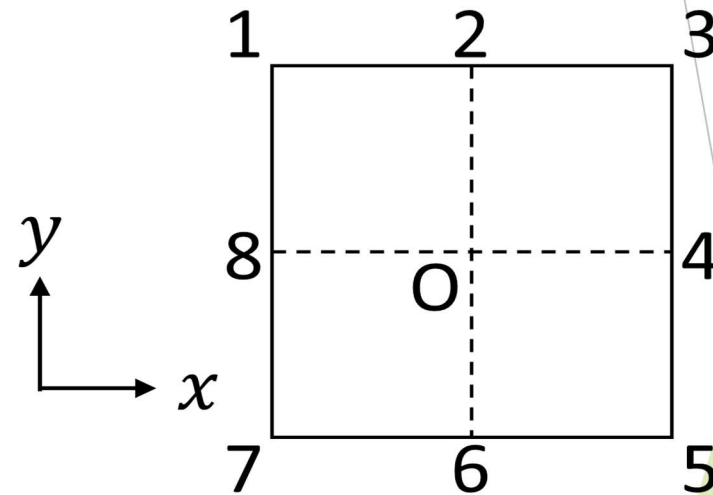
- Two point charges are separated by a distance of 10.0 cm. One has a charge of $-25\ \mu\text{C}$ and the other $+50\ \mu\text{C}$. Determine the direction and magnitude of the electric field at a point P between the two charges that is 2.0 cm from the negative charge?



Answer: $E = 6.3 \times 10^8\ \text{N/C}$.

Example 2:

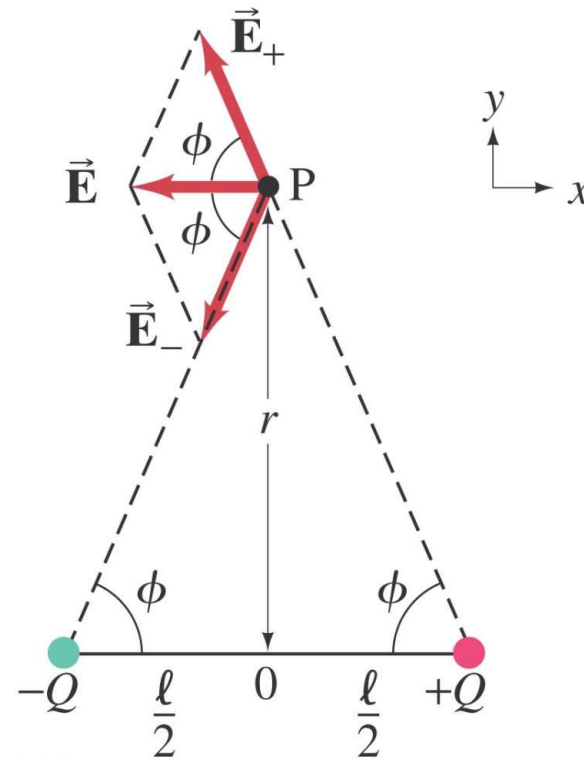
- In the shown figure, the length of the side of the square is $2a$, if $q_1 = Q$ is placed at point “7”, the E at the center of the square is given by ?



$$\text{Answer: } E = \frac{Q}{8\sqrt{2} \pi \epsilon_0 a^2} (i + j)$$

Example 3:

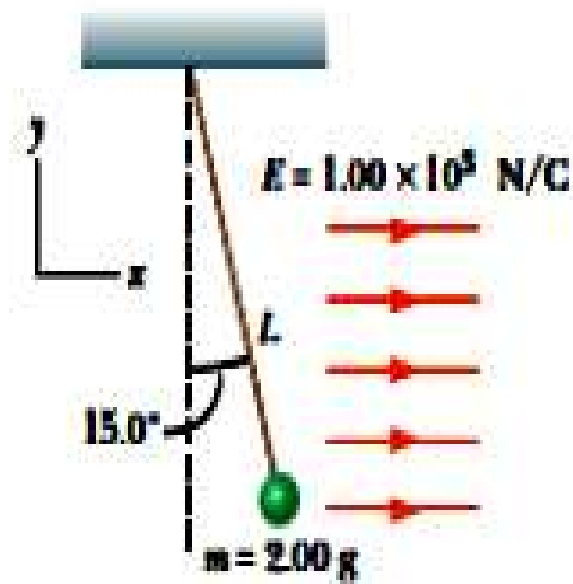
- Charges Q and $-Q$ are located on the x axis, at distance $l/2$ and $l/2$, respectively, from the origin as shown in the Figure. Find the components of the net electric field at P , which is at position $(0, r)$.



$$E = \frac{1}{4\pi\epsilon_0} \frac{p}{(r^2 + \ell^2/4)^{3/2}} \quad \left[\begin{array}{c} \text{on perpendicular bisector} \\ \text{of dipole} \end{array} \right]$$

Example 4:

- ▶ A small, 2 g plastic ball is suspended by a 20 cm long string in a uniform electric field as shown in the figure. If the ball is in equilibrium when the string makes a 15.0° angle with the vertical, what is the net charge on the ball?



Answer: $Q = 5.25$